

Pareto Crewing

One dial, a model-free grade, and price bands:
a decision-theoretic account of choosing language-model teams per task

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Abstract

An agent harness must decide, before each task, which language model sits in which seat: which one writes the code, which one checks it, which one cuts the work into steps. This report gives a decision-theoretic account of that choice and a learner for it. Three commitments distinguish it from the version of the system we shipped. *First*, the reward is a harness-run, model-free grade on the landing — the repository builds, vets, passes the tests of the packages the change touched, and is formatted — yielding one bit $g \in \{0, 1\}$; an off-crew judge’s score and the checker’s finding counts become auxiliary observations whose reliability is fitted, and a run’s own account of itself is never a reward. On 202 real runs we measure what those proxies are worth: a run’s own “settled” claim carries $\rho = 0.02$ of the information of one grade, and an off-crew rating carries $\rho = 0.42$. *Second*, the user-facing control is a single scalar $L \geq 0$, the dollars a failed task costs, and the crew minimises $J(a; x, L) = \mathbb{E}[B(a, x)] + L(1 - P(g=1 \mid a, x))$. We characterise the reachable set: sweeping L over $[0, \infty)$ traces exactly the exposed points of the lower convex hull of the achievable (bill, failure) set, so Pareto-optimal crews strictly inside the hull are chosen by no L — five of twelve on a simulated front. We then prove that the knee our picker ships is always a hull vertex (Proposition 4), and retire it anyway, because its implied stake is never shown to the user and is covariant with the price scale while the user’s stake is not (Proposition 5). *Third*, models are grouped into log-price bands for pruning and for pooled exploration, the crew’s bill is computed from a per-(seat, model) token shape learned from real spend rather than from a fixed default, and one Bernoulli metric `acceptable` carries a source dimension whose per-source reliability is fitted — never two scales in one cell. We give the per-seat additive-logit decomposition and its sample complexity (114 graded runs to separate two models by $\Delta = 0.10$ at a catalog prior worth $\kappa = 30$), the hierarchical catalog-anchored prior and the cap $W = K\ell/(K + \ell)$ derived from an install random effect, variance inflation for drift, the cascade break-even $p_b > c_b/(c_{b'} + (1 - p_{b'})L)$, a switching rule with a cache-loss threshold and an anytime-valid test, and an evaluation protocol. A calibrated simulation reports a negative result we did not want: where every model carries a catalog index, Thompson sampling inside J does not repay its exploration within 800 tasks, and the posterior-mean rule we ship is better; where one model per seat carries no index and is in fact the best, the posterior-mean rule never finds it and its regret does not fall. That asymmetry, not a regret bound, is the argument for a capped and separately-vehicled exploration budget.

Keywords: model selection, decision theory, combinatorial semi-bandits, Thompson sampling, cascades, empirical Bayes, cost-aware learning, LLM agents.

1 Introduction

A harness that drives language models through software tasks makes a choice before every task: which model holds which seat. The choice is made under information that arrives in different units and at different times. A published catalog gives prices and benchmark indexes for every model on the day it appears — 522 models in the feed we read, 301 of them able to call tools, 279 priced and not free. The install’s own past runs are few but on exactly the right distribution. Other installs’ runs are many but on other people’s code.

The system this report describes is the model pool of `codeaf`, and the report has two jobs. The first is to say precisely what the shipped system computes, because a design document that describes an intention rather than a program is worse than none (Section 3). The second is to restructure the decision around an objective that a user can state in dollars, and to be honest about which parts of that restructuring are derived, which are cited, and which are conjecture.

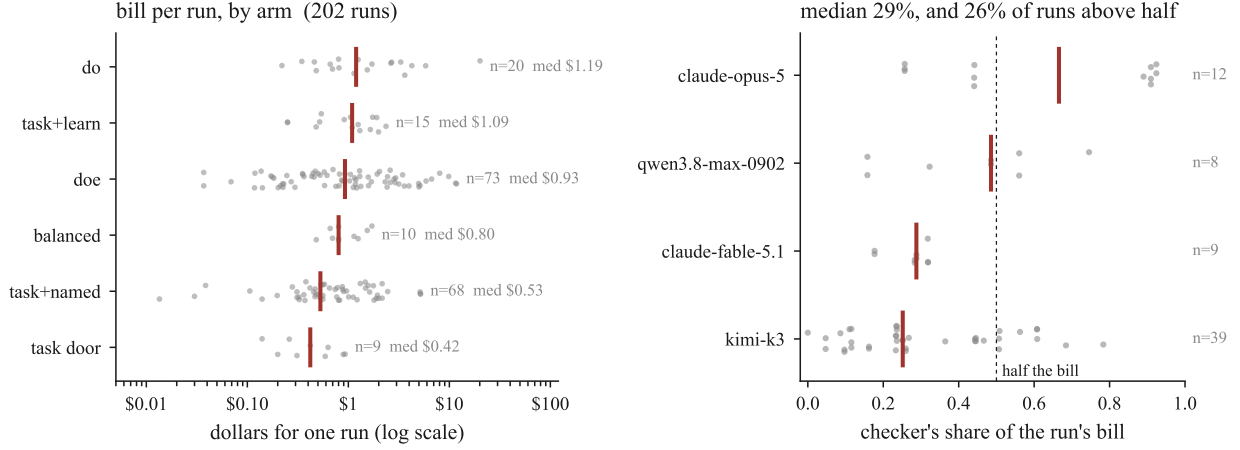


Figure 1: Real data. Left: dollars for one run, by the arm of the design that ran it, over the 202 runs of the internal ledger (one 21-hour window, 2026-09-16/17, one Go repository, internal developers building `codeaf` with `codeaf`). Red bars are medians. Right: the checker’s share of a run’s bill, per checker model, over the 68 runs that carry a per-seat split. A seat that only reads the work takes more than half the bill on a quarter of runs. This is the evidence for Section 8: a picker whose cost model is a fixed default token shape cannot anticipate it.

The defect that prompted this. Under the shipped *learn balanced* rule the picker chose `gpt-5.6-sol` in the worker seat and `fable-5.1` in the checker seat. The checker took 75–80% of each run’s bill and a small task cost \$1.05–\$1.25 against \$0.14–\$0.31 on a crew read from the shipped table. Figure 1 shows the general shape of that from the real ledger: across 68 runs that carry a per-seat split, the checker’s median share is 29%, but 26% of runs put more than half the bill on the seat that only reads the work, and the tail reaches 97%. The picker could not have seen this coming, because its cost model is a fixed default token shape (Section 3), and the knee it optimises is invariant to exactly the thing that would have caught it.

Contributions.

1. A faithful account of the shipped estimator and picker, with the four defects it makes structurally invisible (Section 3).
2. A model-free grade as the reward, with a derivation of what a proxy observation is worth (Proposition 1) and a measurement of three candidate proxies on real runs (Section 4).
3. The one-dial objective, a complete characterisation of what sweeping the dial can and cannot reach (Propositions 2–3), a proof that the shipped knee is always reachable (Proposition 4), and the scale argument for retiring it anyway (Proposition 5); Section 5.
4. Price bands for pruning and pooling, and a learned token shape in place of the default (Sections 6–8).
5. The quality model: additive logit with semi-bandit feedback, a hierarchical catalog-anchored prior, sample complexity in closed form, and drift by variance inflation (Section 7).
6. Exploration, cascades, switching and federation as separate mechanisms with separate proofs and separate limits (Sections 9–12).
7. An evaluation protocol that says what would have to be logged before any off-policy claim is admissible (Section 14), and a limitations section that says what one repository and one day of data cannot support (Section 16).

2 Setting and notation

Definition 1 (Seats, models, crews). Let $\mathcal{S}$ be a finite set of *seats* and $\mathcal{M}$ a finite set of *models*. An *action* (a *crew*) is a map $a : \mathcal{S} \rightarrow \mathcal{M}$ subject to feasibility: each seat s carries a predicate $\text{fit}_s(m)$ (tools, images, context window) and a set $D \subseteq \mathcal{S} \times \mathcal{S}$ of seat pairs that may not be held by models of the same vendor. Write $\mathcal{A}$ for the feasible set.

Our instance has $\mathcal{S} = \{\text{worker}, \text{high}, \text{mastermind}\}$ in the picker: the worker does the work, *high* reads it (we call it the checker in prose, and it is spelled `high` in the code and on the wire), the mastermind cuts the work into steps. A

fourth routing role, **probe**, buys a one-token measurement and is not a crew seat; the judge and the conversational seat are billing seats, not picked seats. Nothing below depends on $|\mathcal{S}| = 3$.

A *task* carries a context x : its size, the door it came through, and whatever else is recorded before the run starts. A *run* is one attempt by one crew on one task. A run’s *landing* is the working tree it leaves behind.

symbol	meaning
$s \in \mathcal{S}, m \in \mathcal{M}$	a seat; a model
$a \in \mathcal{A}, a(s)$	a crew; the model it puts in seat s
x	the task’s pre-run context (size bucket, door, and any other logged covariate)
$g \in \{0, 1\}$	the grade: 1 iff the landing is <i>acceptable</i> by the harness’s own checks
z	an auxiliary observation (judge score, checker findings, a run’s own claim)
$B(a, x)$	the run’s bill in dollars
$L \geq 0$	the <i>stake</i> : what one failed task costs the user, in dollars
$J(a; x, L)$	the objective, Eq. (5)
$\theta_{s,m}$	seat s ’s contribution to the log-odds of acceptance when held by m
α_x	the task’s own difficulty on the log-odds scale
$\varphi_m \in \mathbb{R}^3$	the catalog’s indexes for m (intelligence, coding, agentic)
$w_s \in \Delta^2$	seat s ’s weighting of those three indexes
κ	the catalog prior’s strength, in graded runs
$\sigma_0^2, \sigma_u^2, \tau^2$	prior variance; between-install variance; drift variance per day
$T_{s,m}$	the token shape of (s, m) : input, cached and output tokens per call, calls per task
π_m, π_m^c, χ_m	prompt, cache-read and completion price per million tokens
$b, \beta(b)$	a log-price band; the set of models in it
ρ_{src}	a source’s reliability: what one of its observations is worth in grades
E	the per-install, per-day cap on exploration spend, in dollars
δ	the switching threshold: cache loss plus hysteresis

Table 1: Notation. Dollars are dollars throughout; there is no utility unit and no exchange rate the user does not set.

3 The system as shipped, and what it cannot see

This section describes the program at commit a9fcf9bc8, not a design.

Catalog quality. internal/crewpick scores a model for a seat as a weighted, pool-normalised blend of three published indexes,

$$\hat{q}_s(m) = 100 \sum_{i=1}^3 w_{s,i} \frac{\varphi_{m,i}}{\max_{m' \in \mathcal{M}} \varphi_{m',i}}, \quad (1)$$

with $w_{\text{worker}} = (0.2, 0.3, 0.5)$, $w_{\text{high}} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $w_{\text{mastermind}} = (0.6, 0, 0.4)$. The maximum is taken over the candidate list handed to the call, so the same model scores differently in a stronger pool. A missing index is imputed from ratios to the indexes the model does publish, over at least three donors, capped at the pool’s maximum; an imputed score loses ties.

Measurements enter as a blend. A cell from the pool index or from the install’s own sheet supplies a mean $\bar{y}_{s,m}$ over N observations, and the seat’s score becomes

$$(1 - w) \hat{q}_s(m) + w \bar{y}_{s,m}, \quad w = \frac{N}{N + \kappa}, \quad \kappa = 30. \quad (2)$$

This is `PriorWeightAt` in the code. It is the posterior mean of a conjugate normal model in which the catalog is a prior worth κ observations; Section 7 makes that reading precise and Proposition 6 generalises it.

Bill from a default shape. Each seat carries a *shape*: an input-to-output ratio (100 for worker and checker, 10 for the mastermind), a cache share (0.75, 0.75, 0) and a share of the task’s tokens (0.90, 0.08, 0.02). A seat’s unit cost is $(1 - \gamma_s)\pi_m + \gamma_s\pi_m^c + \chi_m/\text{InOut}_s$ and the crew’s bill is the volume-weighted sum. Every number in that shape is a constant compiled into the binary. Nothing about it is measured, and in particular nothing in it varies by model: a checker that writes ten times as many tokens as another is priced as though it wrote the same number. Figure 1 is what that omission costs.

Front, and the knee. Crews are sorted by bill and swept for strict improvement in quality, giving the Pareto front $\mathcal{F}$; *frugal* is its cheapest member, *max* its best, and *balanced* the member farthest above the chord of the front in $(\ln B, Q)$:

$$\text{balanced} = \arg \max_{a \in \mathcal{F}} \left[(Q(a) - Q_0) - (Q_1 - Q_0) \frac{\ln B(a) - \ln B_0}{\ln B_1 - \ln B_0} \right]. \quad (3)$$

Rewards, today. Two different quantities are published under one metric name, `role.quality`. Live cells are an off-crew judge’s integer score in $[0, 100]$ per seat; the judge is the cheapest catalog model with coding index at least 60 that can call tools, holds no seat in the crew and does not share the worker’s vendor, with at most three tried and 90 seconds per seat. The seed index and the cell ledger instead carry $100 \cdot \mathbf{1}[\text{no major finding}]$ under a $\text{Beta}(1, 1)$ posterior. A Gaussian mean over a mixture of a 0–100 rubric score and a two-point Bernoulli is not a quantity.

Pooling. A row is (role, model, score, judge, door, size, day) under a per-install nonce, with no text, no price and no duration. The relay folds a row into a triple (n, s, s^2) keyed by (install, role, model, judge, door, size, day); copies of a sheet merge by componentwise maximum, which is a join-semilattice, while an install’s own sheet accumulates by addition. The relay fits one additive severity per judge by five alternating weighted-mean sweeps, takes per-install means, and applies a Huber M -estimate (MAD scale 1.4826, tuning constant 1.345, 20 iterations) once a cell has at least five installs, an n -weighted mean below that.

The four things this cannot see.

- G1** *No exploration.* The pick is the argmin at the posterior mean. A seat the pool never rated keeps its catalog score forever. Section 9 shows this is not merely a theoretical gap: it is the one regime in which the shipped rule’s regret does not fall at all.
- G2** *No measured cost.* The token shape is a constant; the bill of a crew is therefore a statement about prices, not about what the models actually do (Figure 1, Section 8).
- G3** *Two scales under one name.* See above; Section 4 separates them.
- G4** *Leakage in the covariates.* The size bucket is computed from the landing’s token count *after* the run ($S < 200k$, $M < 1M$, else L). A feature that is only known after the outcome cannot be conditioned on before the pick. Section 16 says what must replace it.

A fifth, smaller disagreement: the client drops a published cell when its *observation* count is below `min_installs`, while the relay’s floor counts *installs*, and the relay does put an `installs` field on the wire that the client silently discards because it is not a declared string dimension.

4 The reward: a model-free grade, and what a proxy is worth

Definition 2 (The grade). For a landing ℓ , the harness runs, in order: the build over the whole module; `vet`; the test suites of exactly the packages the change touched, with the cache defeated; and a formatting check. Define $g(\ell) = 1$ if all four succeed and 0 otherwise. The grade is a function of the landing and the repository alone. No language model is consulted, so no model grades itself, and no model’s opinion of a landing can move it.

This is a deliberately narrow reward. It says nothing about whether the change is the change that was asked for; Section 16 treats that as the principal threat to validity. What it buys is a reward that is cheap, reproducible by a stranger, and immune to the single largest hazard in an agent harness — that the thing being optimised is a claim made by the thing being optimised.

4.1 Auxiliary observations and their reliability

Three other signals exist and are not thrown away: an off-crew judge’s score in $[0, 100]$; the checker’s counts of major and minor findings; and the run’s own account of how it stopped. They enter as observations of g with a fitted reliability, never as g .

Proposition 1 (What one noisy binary observation is worth). *Let $g \sim \text{Bernoulli}(p)$ and let $z \in \{0, 1\}$ satisfy $P(z=1 \mid g=1) = a$ and $P(z=0 \mid g=0) = b$ with $\gamma := a + b - 1 > 0$, and $z \perp\!\!\!\perp$ everything else given g . Then*

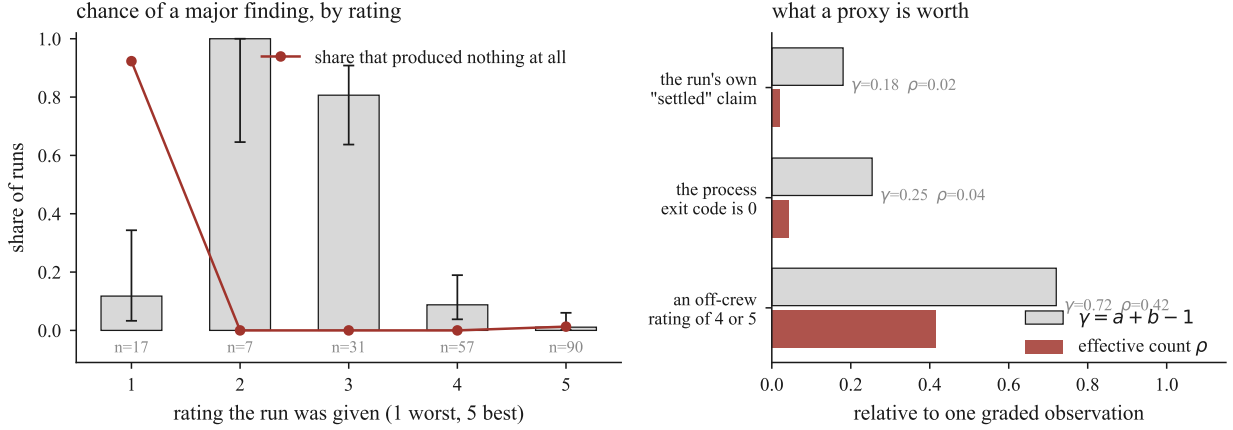


Figure 2: Real data. Left: the chance a run carried at least one major finding, by the rating it was given, with Wilson intervals; the red line is the share of runs at that rating that produced nothing at all. The relation is not monotone, and the non-monotonicity is exactly the runs that did no work. Right: Proposition 1 applied to three candidate proxies for the graded bit. A run's own account of itself is worth $\rho = 0.02$ of a grade.

$z \sim \text{Bernoulli}(q)$ with $q = ap + (1 - b)(1 - p)$, and the Fisher information z carries about p is $\gamma^2/(q(1 - q))$. Relative to one direct observation of g , which carries $1/(p(1 - p))$, the effective count of one z is

$$\rho = \gamma^2 \frac{p(1 - p)}{q(1 - q)} \in [0, 1], \quad (4)$$

with $\rho = 1$ iff $a = b = 1$.

Proof. $q = ap + (1 - b)(1 - p)$ gives $dq/dp = a - (1 - b) = \gamma$. The Fisher information of a Bernoulli(q) observation about its own parameter is $1/(q(1 - q))$; by the chain rule for Fisher information under a smooth reparametrisation, its information about p is $(dq/dp)^2/(q(1 - q)) = \gamma^2/(q(1 - q))$. Dividing by $1/(p(1 - p))$ gives (4). For the range, write $\gamma = a + b - 1$ and note $q(1 - q) \geq \gamma^2 p(1 - p)$ because q is an affine image of p with slope γ into $[0, 1]$: $\text{Var}(z) \geq \text{Var}(\mathbb{E}[z | g]) = \gamma^2 p(1 - p)$, the inequality being the law of total variance with the conditional variance dropped, and it is an equality exactly when $\text{Var}(z | g) = 0$ almost surely, i.e. $a = b = 1$. $\square$

An observation from source src is therefore admitted with weight ρ_{src} in the count N of Eq. (2) and its successors, and $(a_{\text{src}}, b_{\text{src}})$ are fitted wherever grades and proxies co-occur — a Dawid–Skene fit over sources [4], which is the binary case of the many-facet Rasch calibration that the judge-severity fit of Section 3 performs on the continuous scale [17].

4.2 Measured on real runs

Figure 2 applies Eq. (4) to the 202-run ledger, using “the run carried no major finding” as the grade for want of a harness grader in the historical data. Three results.

A run's own claim is worth almost nothing. The harness's settled verdict about its own run has sensitivity 0.56, specificity 0.63, $\gamma = 0.18$ and $\rho = 0.02$: fifty such claims carry the information of one grade. The process exit code does barely better ($\gamma = 0.25$, $\rho = 0.04$). An off-crew rating of 4 or 5 reaches $\gamma = 0.72$, $\rho = 0.42$.

The relation between a rating and a defect is not monotone. At ratings 2 and 3 the chance of at least one major finding is 1.00 and 0.81; at rating 1 it is 0.12. The reason is visible in the cell ledger: 92% of rating-1 runs produced nothing at all — they hit a cap or a provider error with no commit. A run that does nothing has nothing to be wrong about. Any reward monotone in the rating would pay for doing nothing, and any reward monotone in the *absence* of findings would pay for it too. Definition 2 does not, because an empty landing fails the tests the brief asked for.

Therefore the scoreboard splits. Under decision D3 there is one Bernoulli metric `acceptable`, whose cells carry a source dimension in `{grader, judge-major-free, seed}` and whose per-source (a, b) the relay fits alongside the judge severities it already fits; and a separate metric `judge_score` on its own continuous scale. The two are never summed into one cell. This replaces the present arrangement in which a 0–100 rubric score and a two-point indicator are folded into the same (n, s, s^2) triple under the name `role_quality`.

5 The decision: one dial in dollars

Definition 3 (The objective). For a task with context x and a stake $L \geq 0$,

$$J(a; x, L) = \mathbb{E}[B(a, x)] + L(1 - P(g=1 | a, x)), \quad (5)$$

and the pick is $a^*(x, L) = \arg \min_{a \in \mathcal{A}} J(a; x, L)$.

L is the whole user-facing interface: it is the number of dollars a task that has to be redone, or that ships a defect, costs this user. Latency enters, when it is wanted, as $J + \lambda \mathbb{E}[t(a, x)]$ with λ dollars per minute; every statement below survives that addition unchanged, since it only adds a second deterministic term to the bill.

Write $b(a) = \mathbb{E}[B(a, x)]$ and $f(a) = 1 - P(g=1 | a, x)$ at a fixed x , and let $P = \{(b(a), f(a)) : a \in \mathcal{A}\} \subset \mathbb{R}_{\geq 0} \times [0, 1]$ be the achievable set.

Proposition 2 (What the dial can reach). *a minimises $J(\cdot; x, L)$ for some $L \in [0, \infty)$ if and only if $(b(a), f(a))$ is an exposed point of $\text{conv}(P) + \mathbb{R}_{\geq 0}^2$ in a direction $(1, L)$ — equivalently, a vertex of the lower-left convex hull of P . Every such a is Pareto-optimal; the converse fails.*

Proof. $J(a; x, L) = \langle (1, L), (b(a), f(a)) \rangle$ is a linear functional with strictly positive first and non-negative second coefficient. A point of a finite set minimises a linear functional with non-negative coefficients over the set iff it minimises it over the convex hull, and the minimisers of such a functional over a polytope are exactly the faces exposed by that direction; ranging L over $[0, \infty)$ ranges the direction over all of $\{(1, L) : L \geq 0\}$, whose exposed points are the vertices of the lower-left hull. That a minimiser is Pareto-optimal for $L > 0$ follows because a dominating point would attain a strictly smaller value. The converse fails already for three points: with $P = \{(0.1, 0.40), (0.5, 0.30), (1.0, 0.10)\}$ the middle point is Pareto-optimal but lies strictly above the segment joining the other two (which passes through $(0.5, 0.26)$), so no $(1, L)$ exposes it. This is the standard phenomenon of *unsupported* efficient solutions in bi-objective combinatorial optimisation [11, 9]. $\square$

Proposition 3 (The sweep is a finite monotone path). *Order the lower-hull vertices by increasing bill, $a_0, \dots, a_K$. Then $L \mapsto b(a^*(L))$ is non-decreasing and $L \mapsto f(a^*(L))$ is non-increasing; $a^*(L) = a_j$ exactly for L in the interval $[L_j, L_{j+1}]$ with*

$$L_j = \frac{b(a_j) - b(a_{j-1})}{f(a_{j-1}) - f(a_j)}, \quad L_0 = 0, \quad L_{K+1} = \infty, \quad (6)$$

and $L_1 \leq L_2 \leq \dots \leq L_K$. Each L_j is the dollars per unit of failure probability that the j th upgrade buys.

Proof. Consecutive hull vertices a_{j-1}, a_j are indifferent exactly when $b(a_{j-1}) + Lf(a_{j-1}) = b(a_j) + Lf(a_j)$, which rearranges to (6); the hull’s edge slopes $-1/L_j$ increase along the hull by convexity, so the L_j increase, and the argmin steps from a_{j-1} to a_j at L_j and nowhere else. Monotonicity of b and f along the sweep is then monotonicity along the hull, which holds by construction. $\square$

Proposition 3 is what makes the dial implementable: the picker computes the hull once ($O(|\mathcal{A}| \log |\mathcal{A}|)$ after the front), and the answer for any L is a binary search over at most $K + 1$ intervals. It is also what makes the dial explainable: L_j is a sentence — “spending \$0.90 more per task buys one point of acceptance”.

5.1 Why we retire the knee, and why the usual argument for retiring it is wrong

We had expected to show that the knee of Eq. (3) is unreachable. It is not.

Proposition 4 (The knee is always supported). *Let the front have at least two distinct qualities, so the chord’s slope $g = (Q_1 - Q_0)/(\ln B_1 - \ln B_0)$ is strictly positive, and let a_p maximise the bracket of Eq. (3) over the front. Then $(b(a_p), f(a_p))$ is a vertex of the lower-left convex hull of P . Consequently the knee is the argmin of $J(\cdot; L)$ for a non-empty interval of L .*

Proof. Write $D(a) = Q(a) - Q_0 - g(\ln B(a) - \ln B_0)$, so $D(a_p) \geq D(a)$ for all a on the front, and recall $Q = 100(1 - f)$ so that “above the chord in Q ” and “below the chord in f ” are the same statement. Suppose a_p is not a hull vertex. Then there are front points a_u, a_v with $B_u < B_p < B_v$ such that a_p lies strictly above the segment $\overline{a_u a_v}$ in (B, f) , equivalently strictly below it in (B, Q) :

$$Q_p < Q_u + t(Q_v - Q_u), \quad t = \frac{B_p - B_u}{B_v - B_u} \in (0, 1),$$

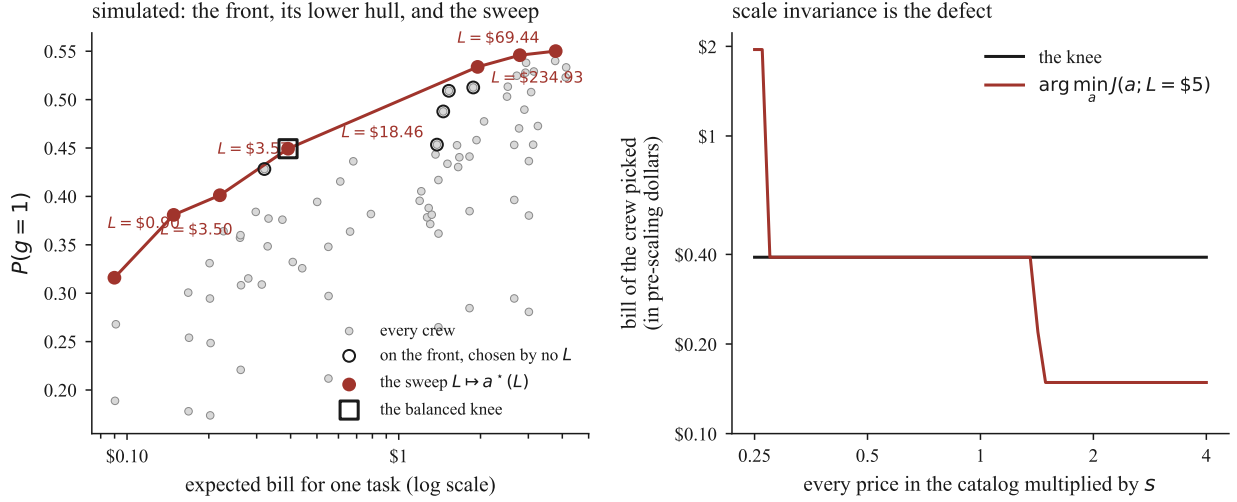


Figure 3: Simulated. A catalog of nine models at each of two seats, prices log-uniform over the span the real ledger shows, quality additive on the logit scale (Assumption 1). Left: all 81 crews; the 12 on the Pareto front; the 7 lower-hull vertices, which are exactly the crews some L selects (Proposition 2), with the breakpoints of Eq. (6); the 5 front crews that no L selects; and the knee of Eq. (3), which is a hull vertex, as Proposition 4 requires. Right: every price multiplied by σ . The knee does not move. The crew that a stake of \$5 selects moves twice.

i.e. $(1-t)(Q_p - Q_u) < t(Q_v - Q_p)$. Optimality of a_p gives $D(a_p) \geq D(a_u)$ and $D(a_p) \geq D(a_v)$, that is

$$Q_p - Q_u \geq g \ln(B_p/B_u), \quad Q_v - Q_p \leq g \ln(B_v/B_p).$$

Combining, $(1-t)g \ln(B_p/B_u) \leq (1-t)(Q_p - Q_u) < t(Q_v - Q_p) \leq tg \ln(B_v/B_p)$, and since $g > 0$,

$$(1-t) \ln(B_p/B_u) < t \ln(B_v/B_p). \quad (7)$$

But $B_p = (1-t)B_u + tB_v$, so by concavity of $\ln$, $\ln B_p \geq (1-t) \ln B_u + t \ln B_v$, which rearranges to exactly $(1-t) \ln(B_p/B_u) \geq t \ln(B_v/B_p)$, contradicting (7). Hence a_p is a hull vertex, and by Proposition 2 some $L \geq 0$ selects it. $\square$

Remark 1. The proof uses only that the knee’s scalarisation is affine in a *concave* increasing transform of the bill. Any such transform — $\ln$, $\sqrt{\cdot}$, a Box–Cox with $\lambda \leq 1$ — gives the same conclusion. A *convex* transform would not, and a knee measured on a convex transform of the bill can indeed be unsupported.

So the knee is a decision. The objection is a different one, and it is fatal.

Proposition 5 (The knee’s stake is covariant with the price scale; the user’s is not). *Multiply every price in the catalog by $\sigma > 0$, so $b \mapsto \sigma b$ and f is unchanged. Then (i) the knee of Eq. (3) is unchanged; (ii) $\arg \min_a \{\sigma b(a) + Lf(a)\} = \arg \min_a \{b(a) + (L/\sigma)f(a)\}$, so the crew that a fixed stake L selects is the crew that the stake L/σ selected before. If the lower hull has at least two vertices, there is an open interval of σ on which the two disagree.*

Proof. (i) $\ln(\sigma B) - \ln(\sigma B_0) = \ln B - \ln B_0$, and likewise for the denominator, so the bracket of (3) is unchanged. (ii) $\sigma b + Lf = \sigma(b + (L/\sigma)f)$ and multiplication by $\sigma > 0$ preserves the argmin. By Proposition 3, $L \mapsto a^*(L)$ is a non-constant step function whenever $K \geq 1$, so $a^*(L/\sigma) \neq a^*(L)$ for σ in an interval around any breakpoint ratio. $\square$

This is the whole case. Prices in this market move by factors of two in months. Under a uniform halving of prices the correct decision moves *up* the front, because a fixed number of dollars of stake now buys twice as much quality; the knee does not move at all, because scale-invariance is what it was designed for. The right panel of Figure 3 draws that: the knee is a horizontal line, and the $L = \$5$ pick is a staircase crossing it. Worse, the stake the knee *implies* is never shown: on the simulated front of Figure 3 the knee sits in the interval $L \in [\$3.57, \$18.46]$, a range in which the user was never asked to have an opinion. The three named presets become three values of L (Section 15), which is the same interface with the units restored.

6 Price bands

Definition 4 (Band). Fix a base $\beta > 1$ and a reference price $\pi_{\min}$. Model m ’s band at seat s is $\text{band}_s(m) = \lfloor \log_\beta (c_{s,m}/\pi_{\min}) \rfloor$, where $c_{s,m}$ is the seat’s expected bill for one task under the token shape of Section 8. We use $\beta = 2$: a band is a doubling of the bill.

Bands do two jobs and neither of them is the decision.

Pruning. $|\mathcal{A}|$ is a product over seats. With 279 priced models the full crew set is far too large to score, and the shipped code already prunes by an 80% quality floor. A band-aware shortlist keeps at most k models per band per seat — the k with the largest posterior mean of $\theta_{s,m}$ — which bounds $|\mathcal{A}|$ at $(k \cdot \#\text{bands})^{|S|}$ and, unlike a quality floor, cannot delete a whole price region. We use $k = 3$. The rule is a heuristic and we do not claim it preserves the hull; a crew whose only virtue is that it is the fourth-best in its band is lost.

Pooling exploration. A model with no observations borrows, through the hierarchy of Section 7, from its band: $\theta_{s,m} \sim \mathcal{N}(w_s^\top \varphi_m + \mu_{s,\text{band}(m)}, \sigma_0^2)$ with a band offset $\mu_{s,b}$ itself drawn from $\mathcal{N}(0, \sigma_b^2)$. Because bills within a band differ by less than a factor of β , the exploration budget of Section 9 can be spent on any member of a band at a known worst-case cost, which is what makes a hard dollar cap enforceable in advance rather than in arrears.

What bands are not. They are not the decision. A band is a bin over one coordinate of P ; the decision is Proposition 2. Reporting “you are on band 4” instead of a stake would reintroduce exactly the unit-free knob we have just retired.

7 Quality: per-seat decomposition, prior, sample cost, drift

Assumption 1 (Additive logit).

$$\text{logit } P(g=1 \mid a, x) = \alpha_x + \sum_{s \in S} \theta_{s,a(s)}. \quad (8)$$

What it buys. The unknowns are $\sum_s |\mathcal{M}_s|$ numbers rather than $|\mathcal{A}|$. Feedback is *semi-bandit*: the grade of one run is one observation for every seat on that crew, so a run informs every crew that shares a model with it. This is the structure of combinatorial bandits with semi-bandit feedback [2, 26], and it is the difference between learning 30 numbers and learning 27,000.

What it ignores. Eq. (8) has no interaction. The interaction we most expect is real and has a name: a strong checker rescuing a weak worker, so that $P(g=1)$ for (cheap worker, dear checker) exceeds the additive prediction. We do not fit it as a parameter. We capture the largest part of it structurally, by the cascade of Section 10, which is an explicit two-stage action whose value is computed rather than learned. A pair term $\delta_{(m_s, m_{s'})} \sim \mathcal{N}(0, \tau_{ss'}^2)$ can be added as a further level of the hierarchy below and costs observations only where a pair is actually run; we regard it as future work and flag any claim about its size as conjecture.

7.1 The prior, and the three-source posterior

Assumption 2 (Hierarchy). With φ_m the catalog indexes and w_s the seat’s weighting,

$$\theta_{s,m} \sim \mathcal{N}(w_s^\top \varphi_m + \mu_{s,\text{band}(m)}, \sigma_0^2), \quad \theta_{s,m}^{(i)} \mid \theta_{s,m} \sim \mathcal{N}(\theta_{s,m}, \sigma_u^2), \quad (9)$$

where $\theta^{(i)}$ is install i ’s own value and σ_u^2 is the between-install variance. Observations carry variance σ^2 .

Every one of the 279 priced models has a prior mean on the day it appears, which is the cold-start property the catalog is for.

Proposition 6 (Three-source posterior). Under Assumption 2, with $k = \sigma^2/\sigma_0^2$, $\ell = \sigma^2/\sigma_u^2$, $\tilde{n} = n^{\text{pool}}\ell/(\ell+1)$ and $K = k + \tilde{n}$, the posterior of install i ’s $\theta_{s,m}^{(i)}$ given the catalog, the pool and n^i own observations is normal with

$$\mathbb{E}[\theta_{s,m}^{(i)} \mid \cdot] = \frac{k^* \hat{\theta}^{\text{cat}} + n^* \bar{y}^{\text{pool}} + n^i \bar{y}^i}{k^* + n^* + n^i}, \quad \text{Var}[\theta_{s,m}^{(i)} \mid \cdot] = \frac{\sigma^2}{k^* + n^* + n^i}, \quad (10)$$

where everything external to this install shares one capped count

$$W = \frac{K\ell}{K+\ell} \leq \ell, \quad k^* = \frac{k}{K}W, \quad n^* = \frac{\tilde{n}}{K}W. \quad (11)$$

Proof. Integrate out $\theta_{s,m}$. The catalog and the pool inform $\theta_{s,m}^{(i)}$ only through $\theta_{s,m}$, whose posterior given them is normal with precision K/σ^2 and mean the $k : \tilde{n}$ precision-weighted average of $\hat{\theta}^{\text{cat}}$ and $\bar{y}^{\text{pool}}$; each pool observation is another install's, hence carries variance $\sigma^2 + \sigma_u^2$ about $\theta_{s,m}$, which is the deflation $\tilde{n} = n^{\text{pool}}\ell/(\ell + 1)$. Adding the between-install variance σ_u^2 gives, for $\theta^{(i)}$, precision $1/(\sigma^2/K + \sigma_u^2) = W/\sigma^2$ with the same mean. Own observations add precision n^i/σ^2 directly. Precision-weighting the two gives (10), and $k^* + n^* = W$ split in the ratio $k : \tilde{n}$. $\square$

Corollary 1. *With $\sigma_u = 0$ ($\ell = \infty$) and no band offset, Eq. (10) is the plain shrinkage blend $(1 - w)\hat{\theta}^{\text{cat}} + w\bar{y}$ with $w = n/(n + k)$, $n = n^{\text{pool}} + n^i$: exactly Eq. (2), with $\kappa = k = 30$.*

Eq. (11) is the point of the hierarchy: however many rows the pool holds, everything external to this install is worth at most $\ell = \sigma^2/\sigma_u^2$ of this install's own observations, because other installs' tasks are not this install's tasks; own observations are never capped. Section 12 gives ℓ its estimator.

7.2 How many runs it costs to tell two models apart

Proposition 7 (Sample complexity at one seat). *Let two models at seat s differ by Δ in $P(g=1)$ at a common operating point $\bar{p}$, let each have κ prior observations and N graded ones, and approximate the grade's variance by $\bar{p}(1 - \bar{p})$. Under the conjugate normal approximation the posterior means differ by a normal with variance $2\bar{p}(1 - \bar{p})/(\kappa + N)$, so a two-sided test at level z separates them when*

$$N \geq \frac{2z^2 \bar{p}(1 - \bar{p})}{\Delta^2} - \kappa. \quad (12)$$

Proof. Immediate from the conjugate posterior $\mathcal{N}(\cdot, \sigma^2/(\kappa + N))$ with $\sigma^2 = \bar{p}(1 - \bar{p})$, independence of the two arms, and the definition of a z -level two-sided test. $\square$

At $z = 1.96$ and $\bar{p} = 0.75$ (so $\bar{p}(1 - \bar{p}) = 0.1875$), Eq. (12) gives, per model, at one seat:

Δ	$\kappa = 0$	$\kappa = 10$	$\kappa = 30$	$\kappa = 100$
0.10	144	134	114	44
0.20	36	26	6	0
0.30	16	6	0	0

Remark 2 (The subtraction is a loan, not a gift). The $-\kappa$ in (12) is only a saving if the prior is centred correctly. A prior worth κ observations that is wrong by more than Δ does not shorten the experiment; it lengthens it, because the first κ observations are spent undoing the prior. The shipped $\kappa = 30$ is a constant, not an estimate; Section 12 gives the moment estimator that should replace it, and Section 9 shows what an overconfident κ costs in the one regime where it matters most.

Two corollaries for the system. First, $\Delta = 0.10$ is out of reach for a single install: 114 graded runs *per model per seat*, against 202 runs in total across all models in our entire ledger. Federation is not an optimisation; it is the only way the question is answerable at all. Second, semi-bandit feedback is what makes even $\Delta = 0.20$ plausible, because each run pays for $|S|$ of these counters at once.

7.3 Drift

Models change under one identifier. Between observations, inflate the variance:

$$\theta_{s,m,t} = \theta_{s,m,t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \tau^2 \Delta t), \quad V_t = \min(\sigma_0^2, V_{t-1} + \tau^2 \Delta t), \quad (13)$$

which is the state equation of a local-level model [13] and the variance-inflation step of Glicko-2 [12]; the cap at the prior variance is what keeps an arm from becoming *more* uncertain than a model nobody has ever run. Two consequences. An arm nobody has run for a while regains uncertainty and is drawn again by the sampler of Section 9 — forgetting is what makes exploration self-renewing without an exploration rate. And a model identifier that is re-released is a change point: the mean and the variance both reset to the prior, rather than drifting there. Figure 5 draws both. τ^2 is estimated by the day-to-day moment $\hat{\tau}^2 = [\text{Var}(\bar{y}_d - \bar{y}_{d-1}) - \sigma^2(1/n_d + 1/n_{d-1})]_+$ pooled over cells, which is why the day is a cell dimension; a cell with no evidence of drift gets $\hat{\tau}^2 = 0$ and forgets nothing. The non-stationary bandit literature [10] offers discounted and sliding-window alternatives; we prefer variance inflation because it leaves the posterior mean alone and therefore does not throw away information that has not been contradicted.

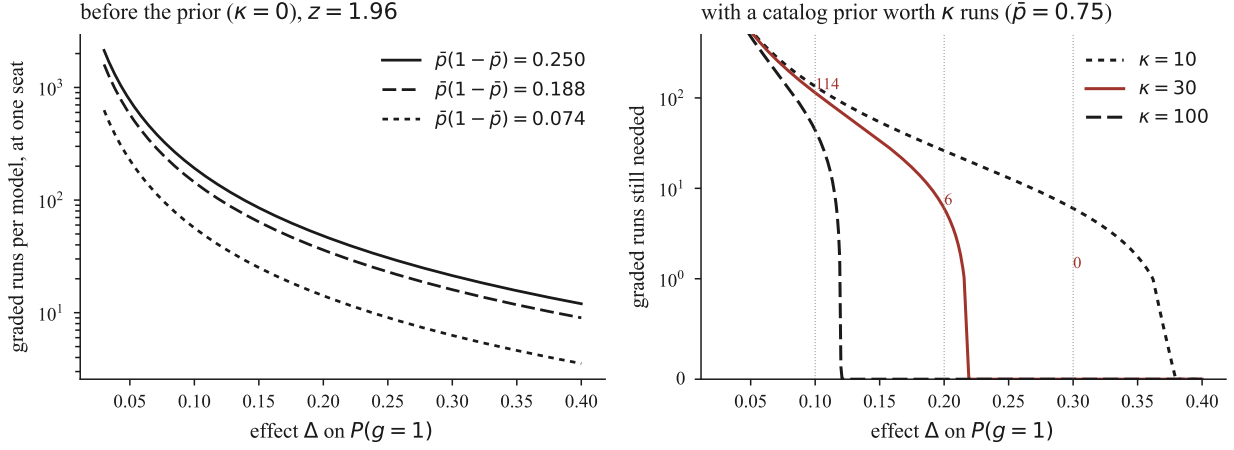


Figure 4: Simulated (closed form). Eq. (12). Left: graded runs per model at one seat before any prior, for three operating points; $\bar{p}(1 - \bar{p}) = 0.074$ is the major-free rate our cell ledger shows, 0.25 the worst case. Right: what remains after a catalog prior worth κ runs, at $\bar{p} = 0.75$, with the $\kappa = 30$ the picker ships marked. Remark 2 applies to every curve on the right.

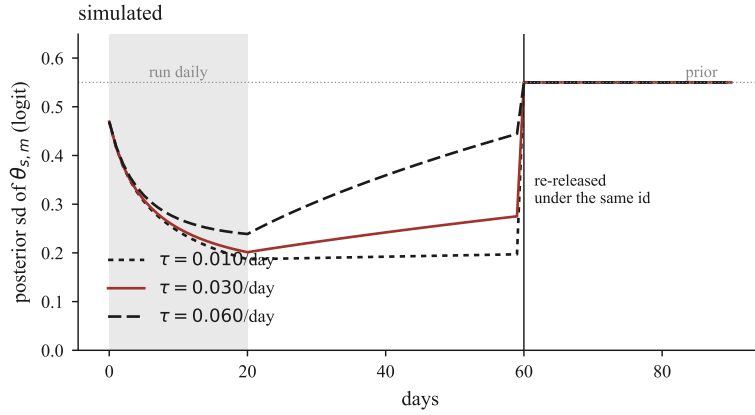


Figure 5: Simulated. Eq. (13). An arm run daily for three weeks, then never; its posterior standard deviation climbs back toward the prior at a rate set by τ , which is what makes it eligible to be drawn again. At day 60 the identifier is re-released and the arm is reset rather than drifted.

8 Cost: a learned token shape

Definition 5 (Token shape). $T_{s,m} = (\bar{l}_{s,m}, \bar{\gamma}_{s,m}, \bar{o}_{s,m}, \bar{v}_{s,m})$: mean input tokens per call, the fraction of them served from cache, mean output tokens per call, and mean calls per task, for model m in seat s .

The bill is then

$$\mathbb{E}[B(a, x)] = \sum_{s \in S} \bar{v}_{s,a(s)}(x) \left[(1 - \bar{\gamma}) \bar{l} \pi_{a(s)} + \bar{\gamma} \bar{l} \pi_{a(s)}^c + \bar{o} \chi_{a(s)} \right] / 10^6, \quad (14)$$

with the four shape components read at the task's context x . Each is estimated from the install's own usage ledger with a shrinkage prior at the seat's default shape:

$$\hat{T}_{s,m} = \frac{n_{s,m} \bar{T}_{s,m} + \kappa_T T_s^{\text{def}}}{n_{s,m} + \kappa_T}, \quad (15)$$

a model with no history priced exactly as today, a model with history priced as it actually behaves. We take $\kappa_T = 5$ runs, on the ground that token shape is far less variable than quality and therefore needs far fewer observations; this is a judgement, not an estimate, and it should be replaced by the between-model moment estimator as soon as there are enough models with history to compute one.

The hypothesis this tests. We state it as such. *Conjecture: the shipped picker's choice of an expensive checker was driven by the default shape's assumption that the checker consumes 8% of a task's tokens, and the actual*

share for verbose checkers is several times that. Figure 1 is consistent with it — checker shares of 63% (median, claude-opus-5) and 44% (qwen3.8-max) against a design assumption of 8% — but the ledger does not record per-seat token counts, only per-seat dollars, so the decomposition into “more tokens” and “dearer tokens” is not identified from the data we have. Eq. (15) is the instrument that would identify it, and logging $T_{s,m}$ per run is the first thing to implement.

9 Exploration

The pick under Eq. (5) at posterior means never tries a crew the prior underrates. Thompson sampling [24, 21] draws one plausible world and acts as if it were true.

Algorithm 1 One pick

Require: task context x ; stake L ; per-seat shortlists S_s (at most k per band, Section 6); the day’s remaining exploration budget E_{left} ; the incumbent crew a_0

- 1: **for** each seat s and $m \in S_s$ **do**
- 2: $\tilde{\theta}_{s,m} \sim \mathcal{N}(\mu_{s,m}, V_{s,m})$ ▷ one draw per task; never redrawn inside a task
- 3: **end for**
- 4: $\tilde{a} \leftarrow \arg \min_a \mathbb{E}[B(a, x)] + L(1 - \text{logit}^{-1}(\alpha_x + \sum_s \tilde{\theta}_{s,a(s)}))$ ▷ costs are *not* sampled
- 5: $\hat{a} \leftarrow$ the same argmin at the posterior means μ ▷ the greedy crew
- 6: $e \leftarrow [\mathbb{E}[B(\tilde{a}, x)] - \mathbb{E}[B(\hat{a}, x)]]_+$ ▷ what this draw would spend on exploration
- 7: **if** $e > E_{\text{left}}$ **then**
- 8: $\tilde{a} \leftarrow \hat{a}; e \leftarrow 0$ ▷ the hard daily cap
- 9: **end if**
- 10: $\tilde{a} \leftarrow$ the crew in $\{a_0\} \cup \{a : a \text{ differs from } a_0 \text{ in at most one seat}\}$ nearest $\tilde{a}$ in J ▷ one seat moves per task
- 11: **return** $\tilde{a}$ if $J(a_0) - J(\tilde{a}) > \delta$ (Section 11), else a_0 ; $E_{\text{left}} \leftarrow E_{\text{left}} - e$

Four things in Algorithm 1 deserve their own justification.

The draw is inside J , not beside it. Because costs are deterministic and L is in dollars, an expensive crew is drawn only when the sampled quality gain covers its extra bill at the user’s own stake. There is no separate exploration rate to set and no unit-free temperature.

A hard daily cap E . Thompson sampling has no budget; J prices exploration but does not bound it. E is a per-install, per-day cap on $\sum_t [B(\tilde{a}_t) - B(\hat{a}_t)]_+$, which makes the worst case a number the user can be told. This is the bandits-with-knapsacks setting [1] in its simplest form — one resource, a hard constraint — and we do not use its algorithms; we use the constraint, and we say plainly that greedily spending the budget until it is exhausted is not the optimal way to spend it.

One seat changes per task. Changing two seats at once makes the resulting grade uninformative about which change caused it, and makes bills jump. The restriction costs optimality — the reachable set in one step is a neighbourhood, not $\mathcal{A}$ — and buys attribution and stability.

Cheap exploration has its own vehicle. The probe role buys a one-token measurement; the cell ledger’s practice of reserving a slot for the least-observed pair is the same idea at the task level. Observations from a deliberately exploratory slot carry a source tag and hence, by Proposition 1, their own reliability; they are not the same evidence as a run the picker chose on its merits.

9.1 What the simulation says, including the part we did not want

Figure 6 is a calibrated simulation: two seats, ten models each, bills log-uniform over the two orders of magnitude the real ledger spans, quality additive on the probability scale, grades Bernoulli, estimation the exact Gaussian linear model (so one grade updates both seats — the semi-bandit structure), prior the truth plus noise held at strength κ .

Where every model carries a catalog index, exploration does not repay itself. Over 800 tasks the posterior-mean rule accumulates \$195 of regret, Thompson inside J accumulates \$347, and the capped version \$270. Thompson’s instantaneous regret falls below the greedy rule’s only at around $T \approx 2000$ in a longer run, and the cumulative debt is not repaid within any horizon a single install will see. The reason is not subtle: a Bernoulli grade has standard deviation near 0.45 about a quantity whose spread across models is near 0.15, so evidence arrives slowly, while $L = \$5$ makes a quality uncertainty of 0.04 worth \$0.20 — more than the whole bill of a cheap crew. We checked the sensitivity: at $L \in \{\$1, \$5, \$20, \$80\}$ the ratio of Thompson’s regret to the greedy rule’s is 1.36, 1.51, 1.70, 1.73. The finding is robust in the direction we did not want.

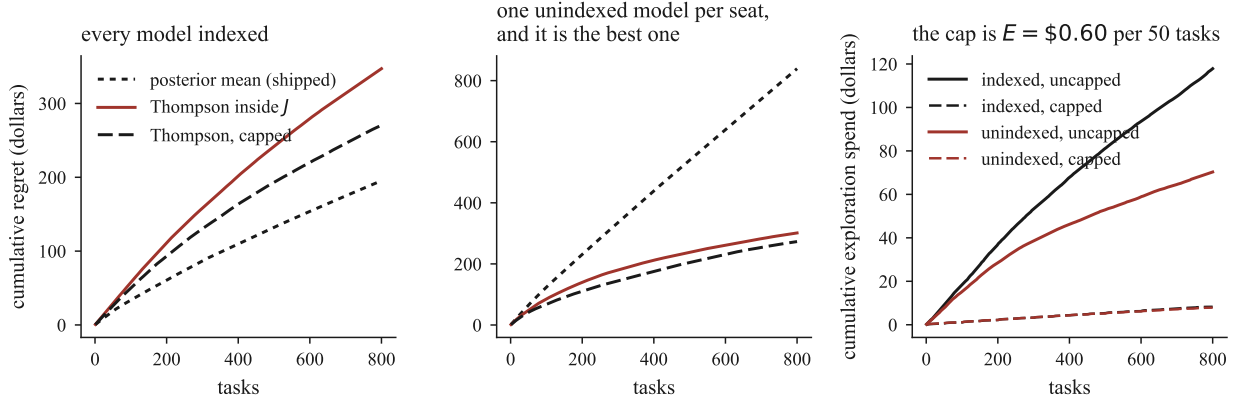


Figure 6: Simulated, calibrated to the ledger’s cost spread. Left: with every model indexed, the posterior-mean rule the picker ships beats Thompson sampling inside J over any horizon an install will see. Middle: with one unindexed model per seat that happens to be the best, the posterior-mean rule’s regret does not fall at all. Right: what exploration costs, and what the per-day cap holds it to. The design conclusion is the asymmetry, not either panel alone.

Where one model per seat carries no index, the greedy rule fails completely. Give each seat one model the catalog does not index — a new release, an open-weights checkpoint, anything outside the feed — give it a prior equal to the seat’s average with three times the prior variance, and make it in fact the best model at its seat. The posterior-mean rule never plays it: its regret over 800 tasks is \$839 and its per-task regret in the last hundred tasks is \$1.00, not falling. Thompson finds it: \$301 cumulative, \$0.19 per task and still falling. The cap costs almost nothing here (\$273 cumulative regret, \$4 of exploration spend against \$70 uncapped) because the exploration that matters is cheap and early.

The design conclusion. Exploration is not worth paying for in the crew choice at this data scale when the prior covers every arm; it is worth almost any price when the prior does not cover an arm. So: keep the posterior-mean pick as the default, keep the Thompson draw behind a setting, keep the hard cap E small, and direct the budget at arms whose *prior* is uninformative rather than at arms whose posterior is merely wide. That is a value-of-information argument, and it is the argument for IDS [19, 20] or a knowledge-gradient-per-dollar rule over plain Thompson sampling. We do not implement either, for a stated reason: both need a per-candidate information estimate at every pick, and at $|\mathcal{A}|$ in the thousands and $|\mathcal{M}|$ in the hundreds that is a second optimisation inside the first, while the gain over Thompson sampling at T in the hundreds is within the noise of Figure 6. We regard “IDS beats Thompson-in- J on this problem” as a conjecture and have not tested it.

9.2 Regret, as a sketch

Proposition 8 (Sketch). Suppose (i) $P(g=1 | a, x)$ is replaced by its linearisation $v^\top \theta_a$ about the operating point, so that $J(a; L) = b(a) + L(1 - v^\top \theta_a)$ is affine in θ with a known deterministic offset $b(a)$; (ii) the prior is the true generating prior; (iii) grades are conditionally sub-Gaussian with proxy ς . Then the Bayesian regret of Thompson sampling over T tasks satisfies $\mathcal{R}(T) = O(L \varsigma \sqrt{dT \log T})$ with $d = \sum_s |S_s|$, by the general posterior-sampling bound of Russo and Van Roy [18], and the d rather than $|\mathcal{A}|$ is exactly the semi-bandit structure exploited by the combinatorial analyses of Chen, Wang and Yuan [2] and Wen, Kveton and Ashkan [26]; for combinatorial Thompson sampling specifically see Wang and Chen [25].

We label this a sketch and mean it. Assumption (i) is false — the link is a logit and the linearisation error is not controlled; assumption (ii) is false, and Section 9 showed the failure of (ii) is what actually governs the outcome; the band pruning of Section 6 changes $\mathcal{A}$ between rounds, which none of these analyses covers; and the exploration cap makes the policy non-Thompson on the rounds where it binds. The bound is offered as an account of *why* the dimension is d and not $|\mathcal{A}|$, and for nothing else. The empirical statement in Figure 6 is not an instance of it.

10 Cascades

Extend the action set. A *cascade* ($b \rightarrow b'$) runs a crew from band b ; if the grade is 0, it escalates to a crew from band $b' > b$; if that also fails, the stake is paid.

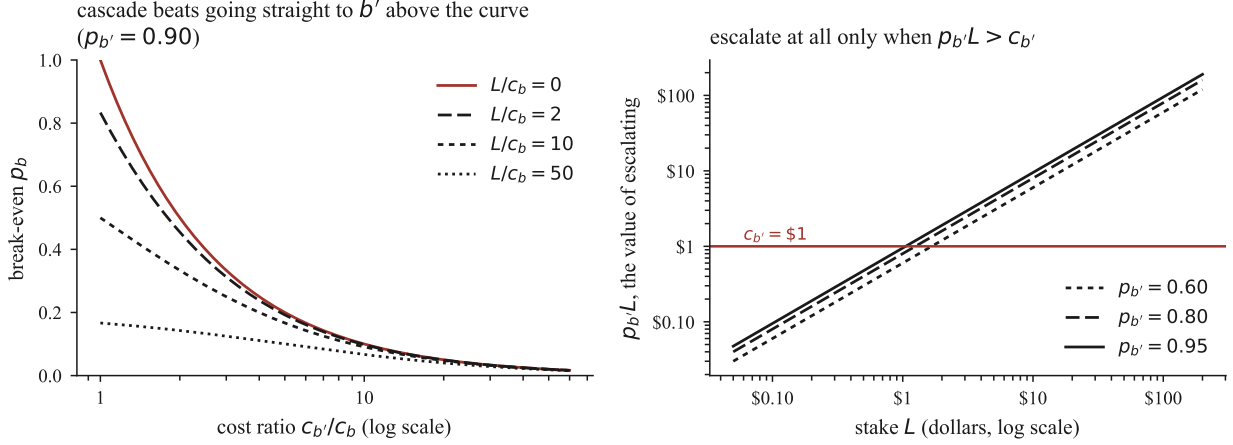


Figure 7: Simulated (closed form). Proposition 9. Left: the break-even acceptance for the cheap band against the cost ratio, for four stakes; $L = 0$ is the bare cost ratio, and every dollar of stake lowers the bar because a cascade strictly reduces the chance of an unaccepted task. Right: condition (18), which decides whether the escalation is worth buying at all.

Proposition 9 (Break-even). Write $c_b, c_{b'}$ for the two bands' expected bills and $p_b, p_{b'}$ for their acceptance probabilities, and let $X = c_{b'} + (1 - p_{b'})L$ be the cost of going straight to b' . Then

$$J(b \rightarrow b') = c_b + (1 - p_b)[c_{b'} + (1 - p_{b'})L], \quad (16)$$

$$J(b \rightarrow b') < J(b') \iff p_b > \frac{c_b}{c_{b'} + (1 - p_{b'})L}, \quad (17)$$

$$J(b \rightarrow b') < J(b) \iff p_{b'}L > c_{b'}. \quad (18)$$

At $L = 0$, (17) is the bare cost ratio $p_b > c_b/c_{b'}$.

Proof. (16) is the expectation over the first grade. Then $J(b \rightarrow b') - J(b') = c_b + (1 - p_b)X - X = c_b - p_b X$, giving (17). And $J(b \rightarrow b') - J(b) = (1 - p_b)[c_{b'} + (1 - p_{b'})L] - (1 - p_b)L = (1 - p_b)[c_{b'} - p_{b'}L]$, which is negative iff $p_{b'}L > c_{b'}$, giving (18). With a latency price λ the same algebra holds with c replaced by $c + \lambda t$ throughout. $\square$

Read together the two conditions say: cascade when the cheap band clears the bar often enough to pay for its own attempt, and when the escalation is worth buying at the user's stake at all. Higher stakes lower the first bar and raise the second, because a cascade ends unaccepted with probability $(1 - p_b)(1 - p_{b'})$, strictly less than $(1 - p_{b'})$: escalation buys quality as well as costing money. Figure 7 draws both. This is the mechanism FrugalGPT studies for single-call inference [3]; the difference here is that the escalation decision is made on a model-free grade rather than on a scored confidence, which is what the next paragraph is about.

The endogeneity hazard, and why D1 removes it. Suppose the escalation trigger were the checker's verdict rather than the grade, and suppose the checker is on the crew. Then a cheap checker that passes bad work produces no escalation, the cheap worker's measured acceptance rises, the picker prefers the cheap crew, and the cheap checker is chosen again. The feedback loop is closed and it collapses to the cheapest possible crew with the worst possible quality, while every measured quantity in the system improves. Formally, the trigger z and the outcome g are both functions of the action a through the checker, so $\mathbb{E}[z | a]$ is not an unbiased signal of $\mathbb{E}[g | a]$ and the argmin of the estimated J is not consistent for the argmin of J . Definition 2 breaks the loop by construction: g is computed from the landing by the harness, and no model on the crew can move it. The judge's score and the checker's findings remain as auxiliary observations with reliability $\rho < 1$, which is exactly the right treatment for a signal that is informative and endogenous: it can shorten the experiment, and it cannot determine its answer.

11 Switching and stability

Definition 6 (Switching threshold). Let a_0 be the incumbent and a_1 the challenger. Switch only at a task boundary, and only when

$$J(a_0) - J(a_1) > \delta, \quad \delta = \underbrace{\hat{\iota}^{\text{cached}}(\pi_{a_0(s)} - \pi_{a_0(s)}^c)}_{\text{cache loss}} / 10^6 + h, \quad (19)$$

where $\hat{\ell}^{\text{cached}}$ is the prompt prefix currently cached for the seat being changed and $h \geq 0$ is a hysteresis margin.

The first term is a real cost with a receipt: changing the model in a seat discards the provider-side cache of that seat’s prompt prefix, and the next call pays the prompt price for tokens that would have been served at the cache-read price. The second is a margin against churn.

Remark 3 ($\delta = 0$ for the worker harness). Every task in the worker harness is a fresh session — there is no cached prefix to lose across tasks — so $\hat{\ell}^{\text{cached}} = 0$ and $\delta = h$. Inside a live chat session the prefix is real and $\delta > 0$; this is why the model in a seat is fixed for the duration of a task and may change only between tasks.

Evaluating at every boundary inflates false switches. The rule of Eq. (19) is tested after every task, which is optional stopping: a fixed- α test applied at every t rejects with probability one under the null. The correct instrument is an anytime-valid confidence sequence: a family $\{C_t\}$ with $P(\exists t : \theta \notin C_t) \leq \alpha$, built from a non-negative supermartingale [15]. Concretely, maintain the sequence for the difference $J(a_0) - J(a_1)$ and switch when its lower bound exceeds δ . The cost is a wider interval than a fixed-sample test at the same α ; the benefit is that “we checked at every task” is no longer a way of manufacturing a switch.

Why batching is the right shape. Bandits with switching costs have minimax regret $\tilde{\Theta}(T^{2/3})$ rather than $\tilde{\Theta}(T^{1/2})$ [5], and the optimal policies batch: they hold an arm for a stretch, then reconsider. Our task boundary is a natural batch edge — it is where the cache is worthless anyway and where a bill is closed — so the switching structure the theory recommends is the structure the system already has. We claim no bound; we claim that the shape is not accidental.

12 Federation

Definition 7 (Cell). A cell is keyed by (seat, model, source, door, size, day) and holds (n, s, s^2) : count, sum and sum of squares, plus the count of contributing installs. A sheet maps (install, cell) to such a triple.

Merging. Two structures, deliberately different, and the previous draft of this document conflated them. *Within* a store, an install’s own sheet accumulates its own disjoint observations by addition, which is associative and commutative but not idempotent, so a retried batch can be double-counted and the relay bounds that with a per-install daily quota. *Between* stores, a copy of another store’s sheet merges by componentwise maximum on each install’s running total, which is a join-semilattice and hence a state-based CRDT [22]: any two nodes that have seen the same publications, in any order and with any duplication, hold the same sheet. The published index is a function of the merged sheet, so a relay is a convenience for bandwidth and signing, not an authority.

What this is and is not. It is centralised-relay and mergeable: one relay publishes, any node can host one, and two relays that receive the same publications agree. It is *not* decentralised learning. There is no coordinated exploration — nothing tells install B that install A has already paid to measure a cell, so the pool re-measures what it already knows. Publishing a per-cell value of information alongside each cell, so that installs allocate their exploration budgets to disjoint questions, is the obvious next step and is exactly the distributed-exploration problem [14]; federated bandit formulations [23] give the communication-regret trade-offs. We have implemented none of it and claim none of it.

The install random effect gives ℓ its meaning. Assumption 2’s $\theta_{s,m}^{(i)} = \theta_{s,m} + u_{i,s,m}$, $u \sim \mathcal{N}(0, \sigma_u^2)$, is what makes $\ell = \sigma^2 / \sigma_u^2$ the cap of Eq. (11): however large the pool, the outside world is worth at most ℓ of this install’s own runs. Every constant is a moment estimate on the sheet, none is set by hand:

$$\begin{aligned} \hat{\sigma}^2 &= \frac{\sum_{i,\theta} (s_{i\theta}^2 - s_{i\theta}^2 / n_{i\theta})}{\sum_{i,\theta} (n_{i\theta} - 1)}, & \hat{\sigma}_u^2 &= \left[\widehat{\text{Var}}_i(\bar{y}_{i\theta}) - \hat{\sigma}^2 \overline{(1/n_{i\theta})} \right]_+, \\ \hat{\sigma}_0^2 &= \left[\widehat{\text{Var}}_{(s,m)}(\bar{y}_{s,m}^{\text{pool}} - \hat{\theta}_{s,m}^{\text{cat}}) - \overline{(\hat{\sigma}^2 + \hat{\sigma}_u^2) / n^{\text{pool}}} \right]_+, & \kappa &= \hat{\sigma}^2 / \hat{\sigma}_0^2, \quad \ell = \hat{\sigma}^2 / \hat{\sigma}_u^2, \end{aligned} \quad (20)$$

each a within-minus-noise estimator of the Efron–Morris kind [8]. Each is withheld until its denominator carries at least ten degrees of freedom, with the day-one values of Table 2 until then. On the seed index alone $\hat{\sigma}_0^2$ clips to zero: thirteen cells of two-point scores cannot resolve a catalog error under about thirteen points, which is why $\kappa = 30$ still stands and why Remark 2 is not an academic caveat.

Robustness and reliability. Rows from one install share a task mix and a codebase; the unit of aggregation is therefore the install mean, not the row. The Huber M -estimate over install means (MAD scale 1.4826, tuning constant 1.345, breakdown 1/2) [16] bounds what a minority of broken or adversarial installs can move, with a precision-weighted mean below five installs. Judge severities are fitted as one additive effect per judge by alternating weighted means, which is the two-way additive model identified up to one constant per connected component of the judge-by-cell graph — the many-facet Rasch calibration [17]. Under decision D3 the same machinery fits one $(a_{\text{src}}, b_{\text{src}})$ per source [4] and publishes ρ_{src} of Eq. (4) in the index, so that every install discounts a proxy observation by the same factor.

constant	meaning	estimator	day-one value
σ	grade / score noise	(20)	— (fitted)
$\kappa = \sigma^2 / \sigma_0^2$	the catalog’s weight, in runs	(20)	30
$\ell = \sigma^2 / \sigma_u^2$	cap on outside evidence	(20)	30
τ^2	drift per day	moment on daily differences	0 (no forgetting)
ρ_{src}	a source’s reliability	(4), fitted per source	grader 1
β, k	band base; keep per band per seat	fixed by design	2, 3
κ_T	shrinkage of the token shape	judgement, see §8	5 runs
L	the stake, in dollars	the user’s	\$1 / \$10 / \$50
λ	dollars per minute	the user’s	\$0.01
E	daily exploration cap	the user’s	a fraction of the daily bill
h	hysteresis in (19)	fixed by design	small

Table 2: Every constant with its estimator. Three belong to the user by design; the rest are empirical-Bayes estimates the relay publishes with the index, or stated design judgements. No constant in this table is a unit-free knob.

13 Architecture

Everything that touches a task runs inside the install: the catalog read, the posterior, the draw, the run, the grade, the own sheet and the outbox. A row leaves the machine carrying no text, no file name, no price and no identity beyond a per-install nonce; dollars and minutes stay in the install’s own ledger. The relay is one edge function over an object store: `POST /v1/rows` folds a row into the sender’s triple and discards the row, so the store’s size is bounded by $\text{installs} \times \text{cells} \times \text{days}$ and a folded row cannot be recovered; `GET /index.json` serves the last published index with a detached Ed25519 signature that the binary verifies locally before reading a single cell. A scheduled job re-fits the severities, the per-source reliabilities and the moment constants of Eq. (20), signs, and mirrors to a public branch. An unreachable relay costs nothing: the cache, then the embedded seed, answer every read, and the outbox waits.

What a compromised relay learns is the sheet — counts and sums per (install nonce, seat, model, source, door, size, day) — and no task, code, prompt, user or address. What it can do is publish a bad index, which the signature stops unless the key is also taken; the minimum-installs floor and the Huber aggregation bound what fabricated nonces can move in the meantime. The floor’s two meanings must be reconciled: today the client counts observations and the relay counts installs, and the relay’s `installs` field is on the wire but silently discarded by the client because it is a number where a declared string dimension was expected.

14 Evaluation protocol

The bar. The system is better if it picks crews that are better on the cost–quality front, per seat and per task, than the shipped posterior-mean blend, on the same data. Not: if it is more principled. Not: if it explores.

Off-policy estimation needs propensities that do not exist. For a logged history $\{(x_t, a_t, g_t)\}$ under a behaviour policy π_0 and a candidate π , the inverse-propensity estimator $\hat{V}_{\text{IPS}}(\pi) = \frac{1}{T} \sum_t \frac{\pi(a_t|x_t)}{\pi_0(a_t|x_t)} r_t$ and its doubly robust refinement [7] both require $\pi_0(a_t | x_t)$ to be recorded at the time of the pick. It is not recorded, and the reason is structural: the crew pick is deterministic, so π_0 is a point mass and every importance weight is 0 or ∞ . The only propensities in the system today are in the router, which records `explore` and `propensity` on the rung-0 attempt of a first attempt — a different decision entirely. *So the first requirement of this protocol is a logging change:* under the Thompson setting, record the sampled $\tilde{\theta}$ ’s seed and the resulting selection probability (estimable by replaying the

draw R times at the stored posterior), and under the greedy setting record that the propensity is one, which makes the honest statement that no off-policy claim is possible from greedy logs.

Until then: replay and simulation, both labelled. Two admissible claims remain. (a) *Replay*: recompute what each rule would have picked for each of the 202 historical tasks and compare the picks’ *bills* exactly — bills are deterministic given the catalog and the shape, so this is an exact statement about cost with no counterfactual outcome in it. (b) *Simulation*, calibrated on the ledger, labelled as simulation in every caption, with its sensitivity to L , to the prior’s error and to the grade’s variance reported, as in Section 9. Neither is an A/B test. An A/B test over crews is the right experiment and is the thing to build next; with $\Delta = 0.20$ and $\kappa = 30$ it needs about six graded runs per model per seat (Proposition 7), which is within reach of a single team over a week.

15 Product semantics

The dial is one sentence: *what does it cost you when a task comes back wrong?* A hobby script, nothing much — a dollar. A change that a colleague will review — ten. A change going near a release — fifty.

preset	L	what it means
<i>frugal</i>	\$1	a failed task is an annoyance; spend nothing to avoid one
<i>balanced</i>	\$10	a failed task costs about ten minutes of a person
<i>max</i>	\$50	a failed task costs an hour of a person, or reaches someone else
<i>custom</i>	any L	the number in dollars

Three properties follow from the machinery rather than from the copy.

Bills become predictable per band. Because the shortlist keeps at most k models per band (Section 6) and the bill is computed from a measured token shape (Section 8), the picker can state, before the run, the band it will run in and therefore the bill’s range within a factor of $\beta = 2$. The old picker could state a bill computed from a constant, which Figure 1 shows was wrong by a large factor at the seat that most often surprises.

Every pick has a reason, and it is the posterior. “Why this crew?” is answered by the two numbers in Eq. (5) and one from Eq. (6): this crew costs $\$x$ and accepts p of the time; the next one up costs $\$y$ more and accepts p' ; at your stake of $\$L$ the upgrade is worth it above $\$L_j$ and you are below it. That sentence is a by-product of the posterior, not a separate explanation feature, and it cannot drift out of agreement with the pick because it is computed from the same numbers.

What exploration costs is a number the user set. E is dollars per day. The system never spends more than E above what the greedy pick would have cost, and it says how much of E it used.

16 Limitations

1. **One repository, one language, one day.** All 202 runs are internal developers building codeaf with codeaf, in one Go repository, inside a single 21-hour window (2026-09-16/17). There are no external installs. Every real number in this document — the checker shares, the proxy reliabilities, the cost spread — is a statement about that sample and about nothing else. The between-install variance σ_u^2 , on which the cap of Eq. (11) entirely depends, is unmeasured, because there is exactly one install.
2. **g is binary and only as good as the tests.** Definition 2 rewards a landing that builds, vets, passes the touched packages’ tests and is formatted. A change that does all four and does not do what the brief asked scores 1. A repository with weak tests grades weakly. We regard this as the dominant threat to validity: the reward is honest and narrow, and the narrowness is not free. Extending g to a graded scale (fraction of acceptance criteria met) is straightforward statistically and hard to specify in a way that no model can influence.
3. **The objective is risk-neutral.** Eq. (5) optimises an expectation. A user who cares about the worst task of the week, not the average, wants a mean-variance or CVaR objective; $J_\eta(a) = \mathbb{E}[B] + L(1 - P) + \eta\sqrt{\text{Var}[B]}$ is a one-line change with a second dial, and a second dial is exactly what Section 5 argued against. We have not resolved this.
4. **The additive logit ignores interactions.** Assumption 1 is a working rule. The cascade captures the largest interaction we expect; a pair term would capture more at a cost in observations we cannot currently afford (Proposition 7).

5. **The size feature leaks.** The size bucket is computed from the landing’s token count after the run (G4). It must be replaced by a pre-run estimate — the brief’s own token count, the number of files named, the door — and until it is, every conditional statement on *size* in the shipped system is contaminated. Nothing in this report uses it.
6. **The judge is the cheapest passable model.** The judge is chosen as the cheapest catalog model above a coding index of 60 that can call tools and holds no seat. That is a budget rule, not a measurement rule, and it makes the judge’s severity correlated with its price. The severity fit removes an additive offset and cannot remove a difference in what a cheap judge is capable of noticing.
7. **Exploration was not shown to pay.** Section 9 reports a negative result. We have not shown that Thompson sampling inside J improves this system in the regime it will mostly run in; we have shown where it is indispensable and where it is a tax, and set the default accordingly.
8. **No off-policy evidence is possible from the logs we have** (Section 14), and none is claimed.
9. $\kappa_T = 5, k = 3, \beta = 2, h$ **are judgements.** They are marked as such in Table 2 and each should be replaced by an estimate.

17 Related work

The decision-theoretic core is bi-objective scalarisation and the classical fact that linear scalarisation reaches exactly the supported efficient set [11, 9]; Pareto bandits [6] address the case where no scalarisation is available, which is the case we deliberately leave by putting the exchange rate in dollars. The learner is a combinatorial bandit with semi-bandit feedback [2, 26, 25] under Thompson sampling [24, 18, 21]; information-directed sampling [19, 20] is the value-of-information alternative discussed in Section 9. The budget is the bandits-with-knapsacks constraint [1] without its algorithms. Non-stationarity is handled by variance inflation in the manner of Glicko-2 [12] and the local-level model [13], rather than by the discounting and sliding windows of [10]. Switching costs and their $T^{2/3}$ consequence are Dekel et al. [5]; anytime-valid testing at every boundary is Howard et al. [15]. Cascades over price tiers for language models are FrugalGPT [3]. Rater calibration is many-facet Rasch [17] on the continuous scale and Dawid–Skene [4] on the binary; shrinkage is Efron–Morris [8] and robustness is Huber [16]. Merging without a coordinator is the state-based CRDT [22]. Distributed and federated bandits [14, 23] describe the coordinated exploration we do not yet do. Off-policy evaluation is Dudík et al. [7].

18 Status

Shipped at the commit this document describes: the catalog picker with the front and the knee; the shrinkage blend of Corollary 1 at $\kappa = 30$; the index document, its reader, its signature and the embedded seed; the outbox, the install nonce and the push; the judge and its deterministic choice; the own sheet; the relay with its store, its join, its judge-severity fit and its Huber aggregation; the mirror.

Specified here and not yet in code: the grade of Definition 2; the single acceptable metric with a source dimension and fitted ρ_{src} ; the stake L and the hull sweep of Propositions 2–3 in place of the knee; the bands of Section 6; the learned token shape of Eq. (15); the drift inflation of Eq. (13); the hierarchical cap of Eq. (11) with the moment estimators of Eq. (20); the Thompson draw, the cap E and the one-seat rule of Algorithm 1; the cascade actions of Section 10; the switching rule of Eq. (19); the propensity logging of Section 14; a pre-run replacement for the size feature.

Reproducibility. The script `extract.py` turns the private run ledger into three text-free CSV files under `data/`, and `make_figures.py` draws every figure from those files and from the simulations described in its docstrings. Both live in `figures/` beside this document and are committed with the CSVs, so every figure rebuilds on a machine that has never seen the ledger. Figures 1 and 2 are real data; Figures 3, 4, 5, 6 and 7 are simulation or closed form, and say so in their captions.

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